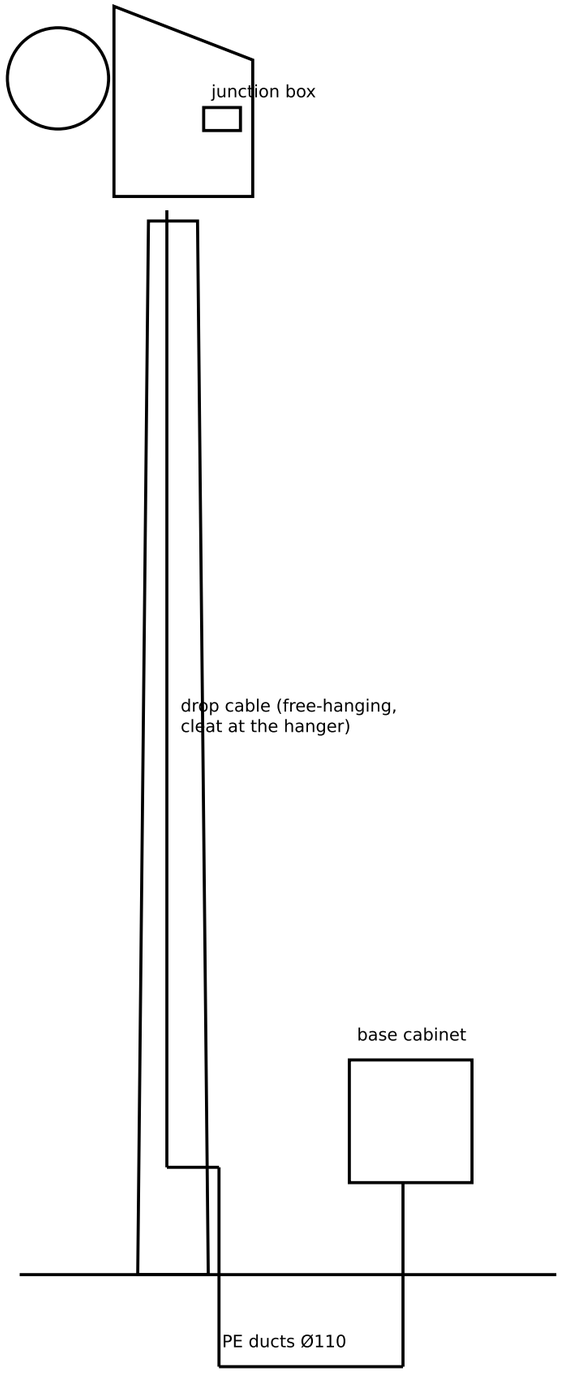
ROUTE (nacelle → loop → tower → cabinet)

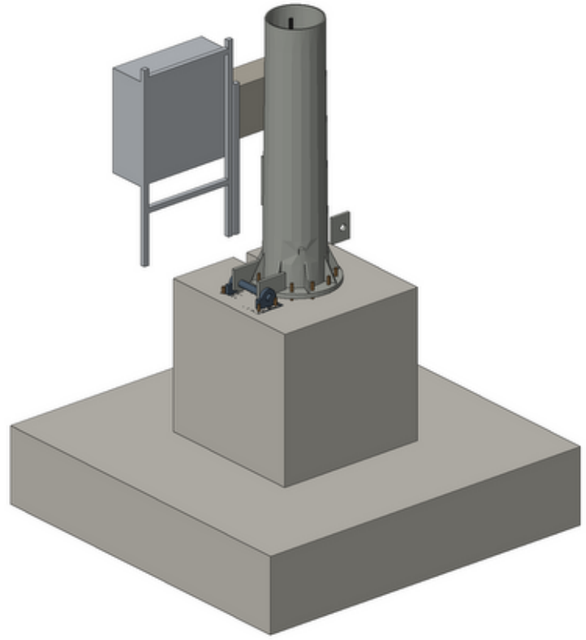


Cable	Type	From → to	Length m
W1 power	H07RN-F 4G10 (3 ph + PE)	generator terminals → junction box	3
W2 power	H07RN-F 4G10, 2.5 turns loop	junction box → hanger loop → hand hole	12
W3 power	NYY-J 4 × 10 in duct	hand hole → cabinet	4
W4 brake	ÖLFLEX 4G1.5 shielded	caliper → junction box → cabinet	16
W5 sensors	LiYCY 12 × 0.5	overspeed, encoder, cam, met, PT100 → cabinet	16
W6 yaw motors	ÖLFLEX 7G1.5	2 gearmotors + brakes → cabinet	16
W7 earth	Cu 50 mm² green/yellow	air terminal → bedplate; tower lug → earth ring	6
W8 data	CAT6 outdoor or fibre	cabinet → house / grid meter	site

NOTES

1. Cable loop NY-460: power and control cables hang free in the tower below the cleat with 2.5 turns of slack; cable-twist limit ±2.5 turns from the encoder/cam switch (doc 25), auto-untwist when idle.
2. Lightning (LPL I, doc 02): air terminal → bedplate → yaw pad brush → adapter → tower → base flange → 2 earthing lugs → ring earth (≤ 10 Ω). Generator winding star point not earthed (IT, insulation monitor in the cabinet).
3. Surge protection: T1+T2 arresters at the cabinet entry for W2/W3, data-line arresters for W5/W8; shields bonded at both ends via the junction box gland plate.
4. All cables UV and cold resistant to −40 °C; drip loops at the hand hole; glands IP68; label every core at both ends.

AXONOMETRIC VIEW (not to scale)



Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

CABLING ROUTES, EARTHING AND LIGHTNING

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-800	B · 2026-11-06	schematic
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see cable schedule	–	Claude / not checked